Homework Assignment 4: Studention Transport Monte Carlo Simulation

Tal Ramot

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1 Introduction

This report details the results of an explicit Monte Carlo simulation of Studention transport in a sphere of Exercisium. Exercisium is a fissile material with atomic weight $A = 300$, density $\rho = 30$ g/cm^3, and a microscopic cross section $\Sigma_t = 10$ barn.

The simulation advanced active Studentions generation-by-generation. For each Studention, the distance to the next event is determined by $s = -\ln(\xi)/\Sigma_{macro}$. If the advanced particle leaves the sphere radius R, it is deleted. Otherwise, the type of collision is determined probabilistically:

- **Absorption** ($P_0 = 20\%$): The Studention disappears.
- **Scattering** ($P_1 = 50\%$): The Studention scatters in a random 3D direction.
- **Fission** ($P_2 = 30\%$): The original is absorbed, and two new Studentions are emitted in independent random directions.

A sphere is considered critical if at least one of 100 histories starting at the origin becomes critical (surpassing 10,000 active particles). Otherwise, the sphere is noncritical.

2 Question 1: Critical Radius of the Base Case

Running the bisection search with $N = 100$ histories per check yielded:

- **Critical Radius:** [Insert Critical Radius] cm
- **Critical Mass:** [Insert Critical Mass] g

3 Question 2: Critical Mass vs. Density

The simulation was executed for 11 densities linearly spaced between 10 g/cm^3 and 100 g/cm^3. The resulting critical radius and critical mass are plotted on a logarithmic scale in Figure 1.

3.1 Discussion

[Write explanation of how critical radius and critical mass depend on the density of the fissile material.]

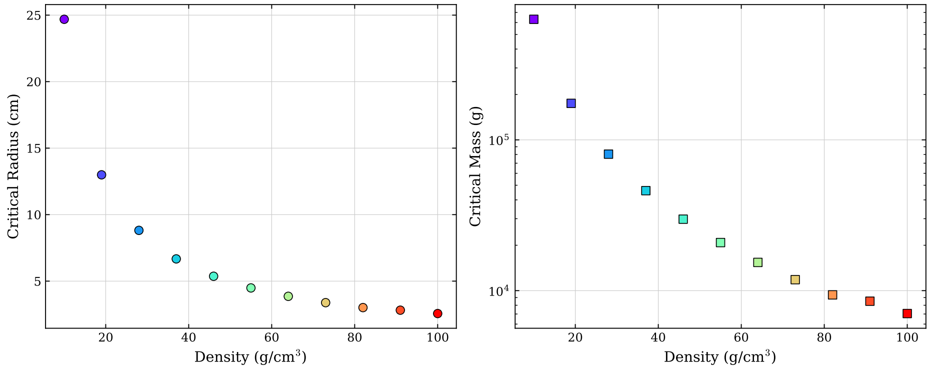

Figure 1: Critical radius (cm) and critical mass (g) as a function of Exercisium density.

4 Question 3: Probability Grid & Parameterization

The critical radius and critical mass were evaluated across 13 different collision probability configurations. The results are plotted against the multiplication parameter $c = P_1 + 2P_2$ in Figure 2.

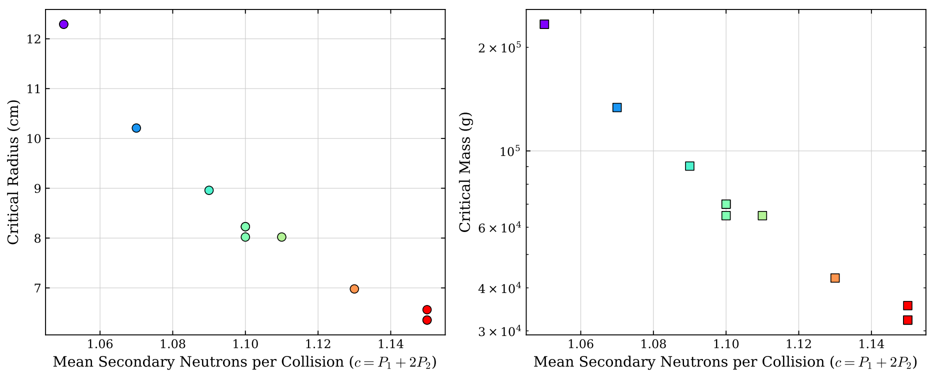

Figure 2: Critical radius (cm) and critical mass (g) as a function of the multiplication parameter $c = P_1 + 2P_2$.

4.1 Discussion

[Discuss whether the critical mass is uniquely determined by the average secondary neutron count c, or if individual P_0, P_1, P_2 values have distinct effects.]

5 Question 4: Deep Probability Sweep

A wider grid of 33 different probability configurations was simulated to analyze the critical radius and mass across a broad range of c. The results are shown in Figure 3.

5.1 Discussion

[Explain whether the results are determined mainly by c, or whether the separate values of P_0, P_1, and P_2 also have a visible effect.]

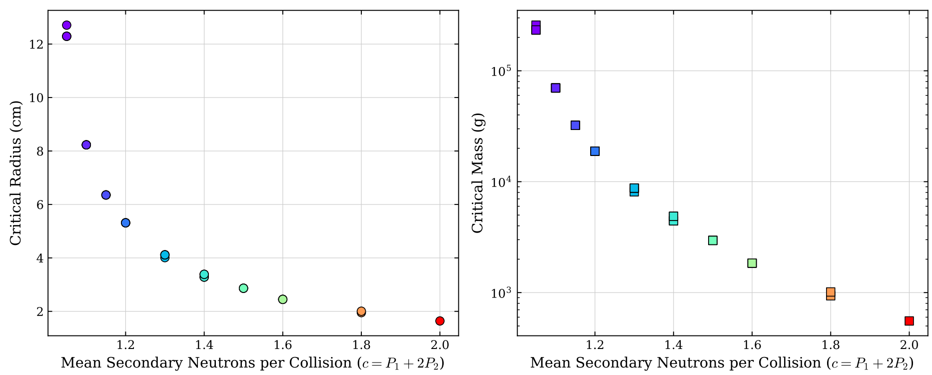

Figure 3: Critical radius (cm) and critical mass (g) for 33 configurations as a function of the multiplication parameter $c = P_1 + 2P_2$.